

Report on
Evolving Reputation for Commitment:
Understanding Inflation and Inflation Expectations
by Robert King and Yang Lu

This paper considers a central banking reputation game.

My main comment is that I am having trouble understanding the setup as a formal game. That is, Players, Strategies, Payoffs. I believe the exposition would be greatly enhanced putting everything this way.

Obviously the players at any given time are the Policymaker and a continuum of private sector agents. The strategies are not so clear to me.

I believe it is the following:

Suppose a public regime change has just happened: Then a new policymaker is drawn and the game essentially starts over. Call this date $t = 0$. With probability ρ_0 it is the commitment type and $(1 - \rho_0)$ it is the opportunistic type. (As a side note, I have trouble figuring out how this can depend on the previous reputation. Reputation isn't a physical thing. I would welcome the authors specifying exactly how the type of the new policymaker at what is essentially the start of a new game depends probabilistically on something real like previous actual actions in the previous play of the game). Next, a shock ς_t is publicly observed. The policymaker then announces a state contingent action, $a_t(h^t)$ for every date $t \geq 0$ where h^t is the history of public observed variables up to and including t . (This is a large object.) After the announcement, things get a bit confusing. The authors say the private sector agents "form inflation expectations". Would it be ok to say they each independently choose an action E_0 ? One could then make payoffs such that the optimal way to choose E_0 is to equal the expected value of something else.

But let's get back to the timing: After E_0 is chosen by each private sector agent, the policymaker privately chooses an action a_0 . If the policymaker is the commitment type, its action set is the singleton of what it announced a_0 would be. If it is the opportunistic type, its action set is (I think) any real number.

But here I get confused again. Equation 2 says a public random variable (called inflation) π_0 is drawn such that

$$\pi_0 = a_0 + v_{\pi,0},$$

where $v_{\pi,0}$ is a normally distributed mean zero random variable. The idea is that a_0 and $v_{\pi,0}$ are private, but π_0 (their sum) is public. But then that makes equation 1 confusing, which appears to say that actual inflation is determined by expectations of future inflation. My best guess is that the authors are building equation 1 into payoffs, but it would be much better for them to be more explicit about this. That is, once you have realized inflation π_0 and E_0 , you use equation 1 to get a variable called the x_0 or the "output gap", and then the payoff to the policymaker depends on π_0 and x_0 . It is here that I would make the payoffs to private agents

$$-(E_0 - \pi_0)^2.$$

For subsequent dates, if the game doesn't start over, the commitment type has to announce what the $t = 0$ plan said he was going to announce, and has to actually take the action $a_t(h^t)$, that the plan says as well. And the opportunistic type can do whatever it wants.

So a *strategy* is a time zero plan $\{a_t(h^t)\}_{t=0}^\infty$ for the commitment type, how the opportunistic type acts, denoted $\{\alpha_t(h^t)\}_{t=0}^\infty$, and how the private sector agents play, or $E_t(h^t)$. An equilibrium is every player optimizing at every node of the game. Reputation comes into play because the private sector agents need to derive the endogenous probability they are facing the commitment type at each node.

But the difficult thing is the plan $\{a_t(h^t)\}_{t=0}^\infty$ itself, and this is where the paper's value comes in. Generally, you would get a huge set of sequential equilibria if the game theorist does not have further discipline on where this comes from. (Pick an arbitrary $\{a_t(h^t)\}_{t=0}^\infty$ and have the private sector agents believe the policymaker must be the opportunistic type if it doesn't announce exactly that plan).

So what the authors do is essentially an equilibrium selection device: the strategy $\{a_t(h^t)\}_{t=0}^\infty$ must solve a mechanism design problem of making the commitment type as well off as possible subject to the other equilibrium conditions holding. This seems like a reasonable device to me if you can pull it off, which the authors do. And having such a way of NOT having a huge set of equilibria then allows the authors to actually compare the predictions to data, and the model does surprisingly well.

So my overall view is that I like this paper, but want the authors to be much more clear about the nuts and bolts of the game while presenting it.